ConfAgent

置信度驱动的智能体递归自我改进

阶段性实验汇报 · 2026-10-20

50 / 198

旧主任务 / 调用，反复重评不重复计数

30 / 50

新增任务 / 调用，冻结校准器验证

阶段一原型

概率输出已接通，纠错收益尚未验证

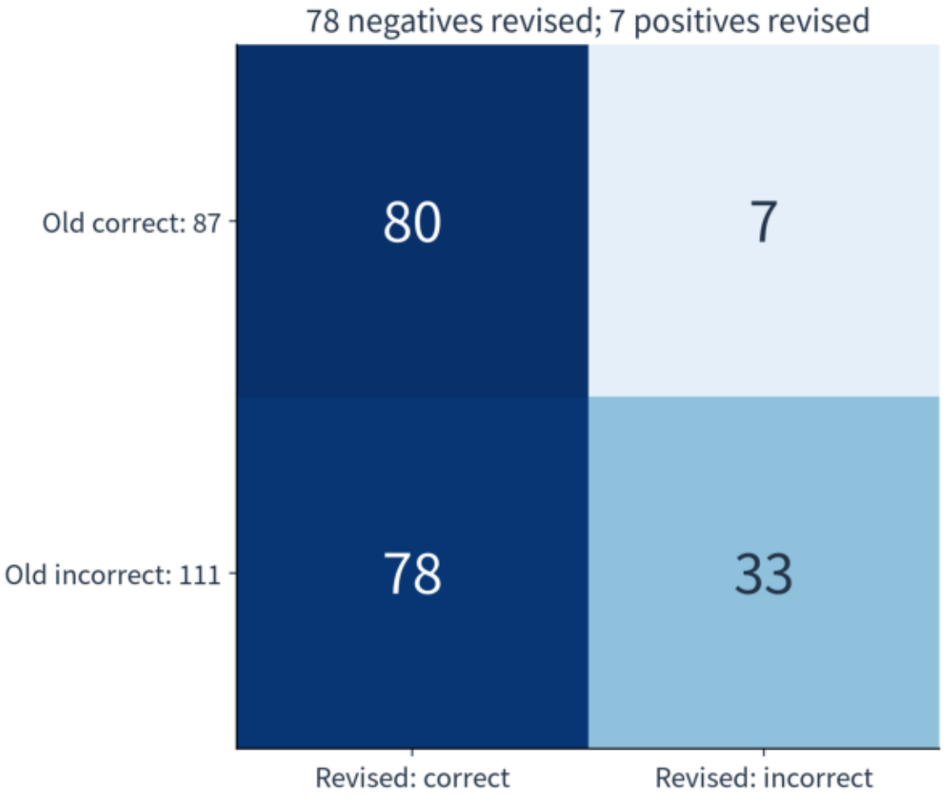
核心发现：标签口径会改变结论；一致性有开发集价值；内部校准改善概率误差。

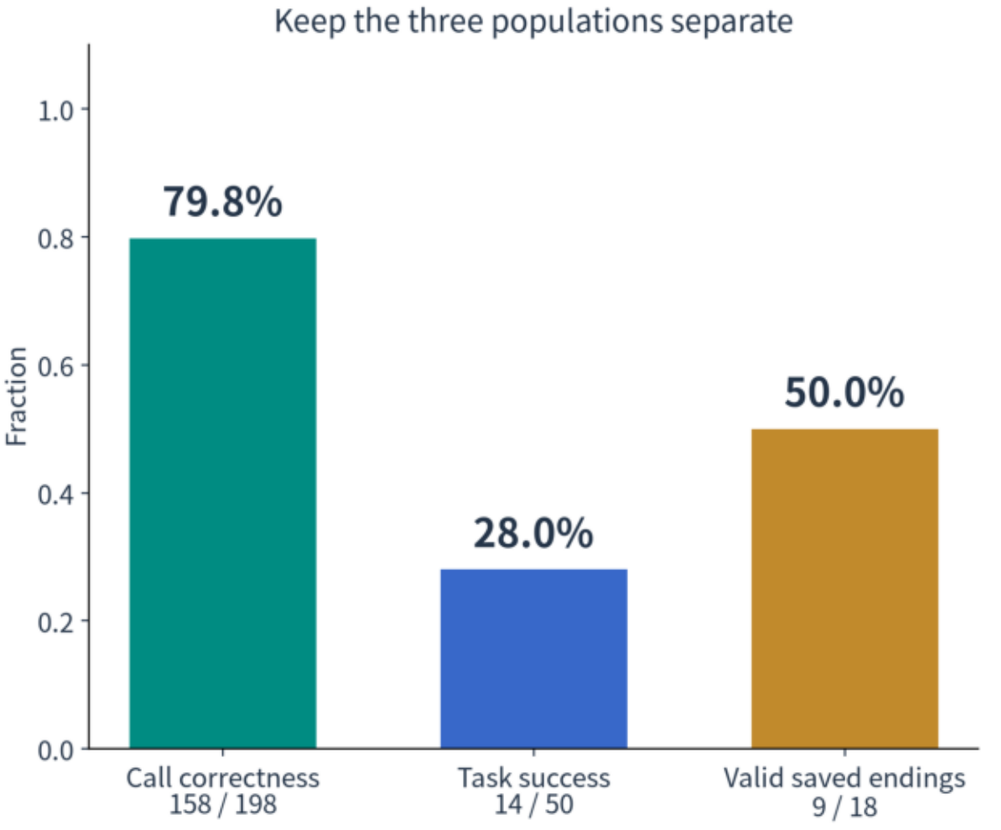
核心限制：新域错误过少，校准未胜常数基线；尚无实际纠错与递归能力提升证据。

全部图表由冻结结果离线生成；未新增模型请求。完整说明、数据和代码在同目录。

01 Relabeling changes measurement, not model ability

Same 50 tasks and 198 calls; 18 saved no-tool outputs retain separate ending labels.





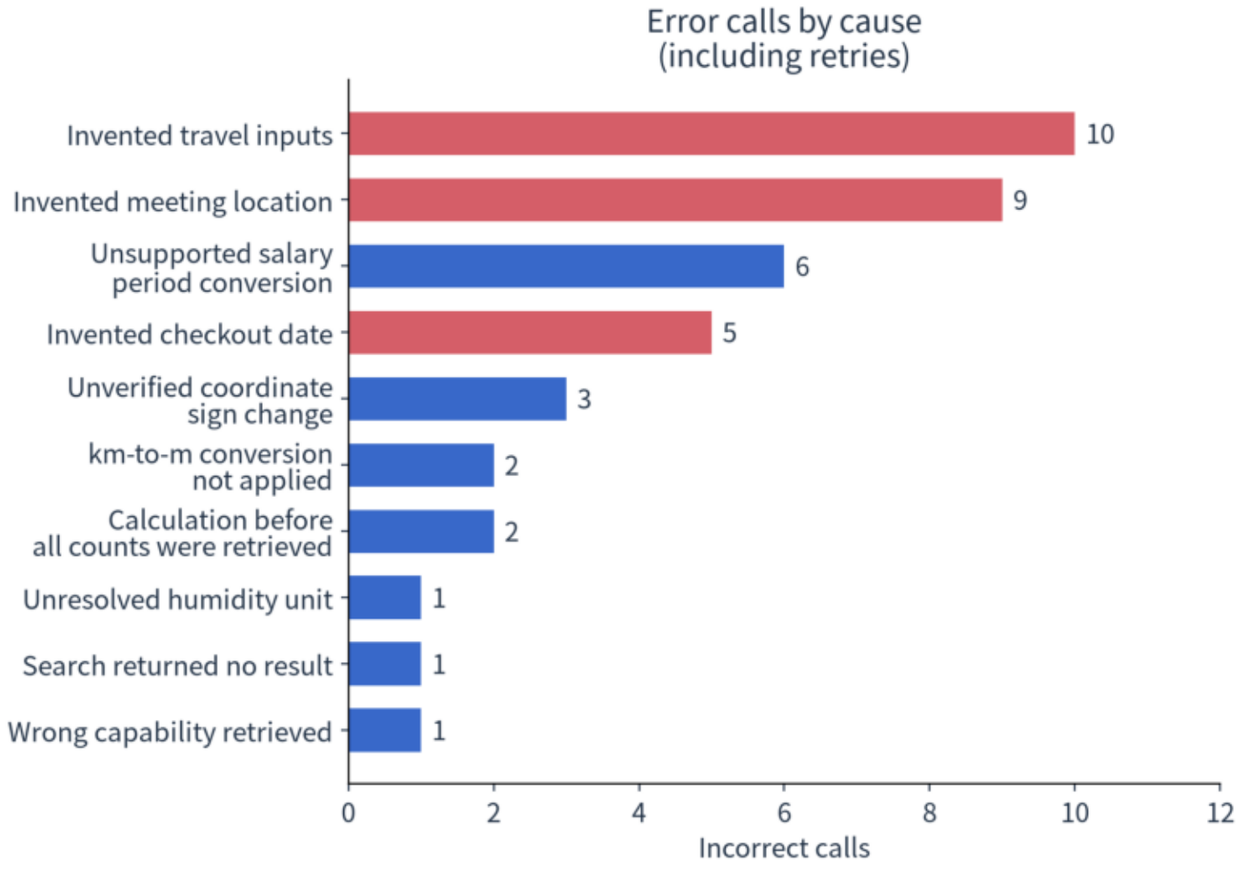
85/198 call labels changed. Logprob AUROC 0.5401 → 0.6351 reflects relabeling; the model was not updated.
Protocol-based Codex review; no independent human double annotation or annotator-agreement study.

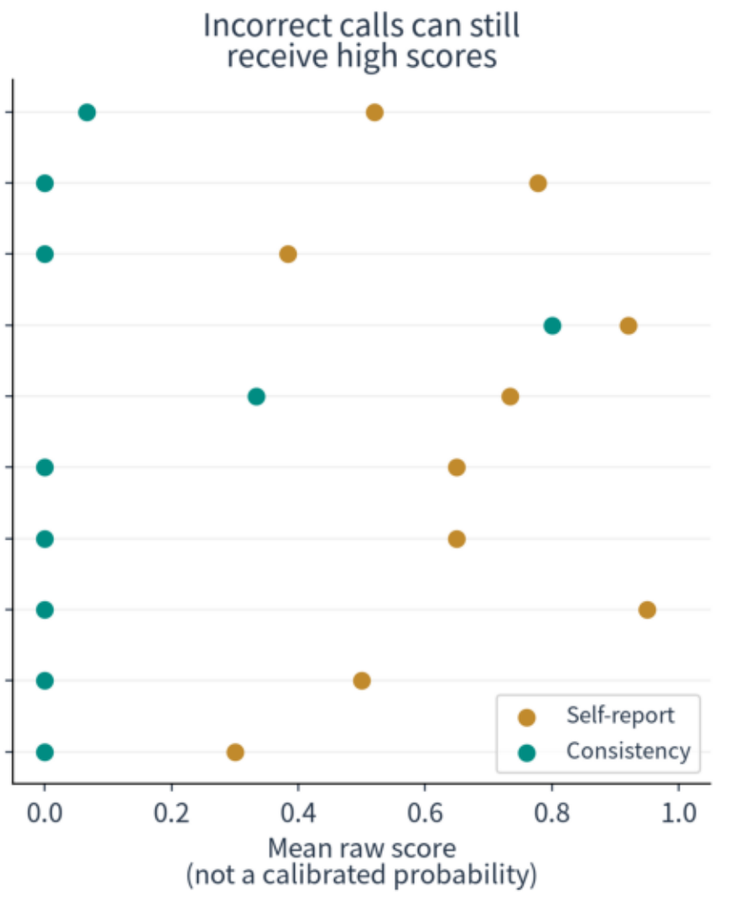
Sources: legacy_binary_v2/label_summary.json; logprob_metrics.json | As of 2026-10-20

85 call labels changed. Evaluate local calls and complete tasks separately.

02 Unsupported inputs dominate the reviewed errors

24 of 40 incorrect calls invent missing travel inputs, meeting locations, or checkout dates.





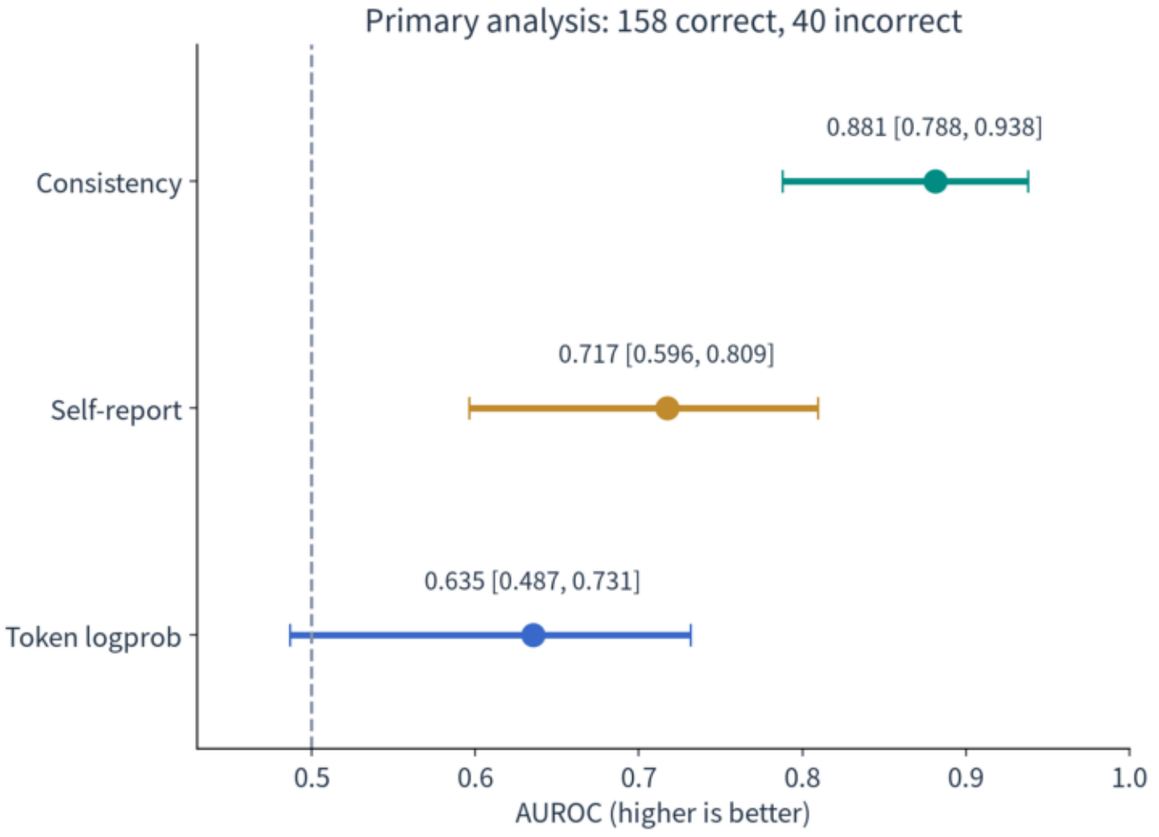
Five invented-checkout errors: mean self-report 0.92 and consistency 0.80. Repeated errors can remain consistent.
The 40 incorrect steps come from 16 tasks; they are not 40 independent error tasks.

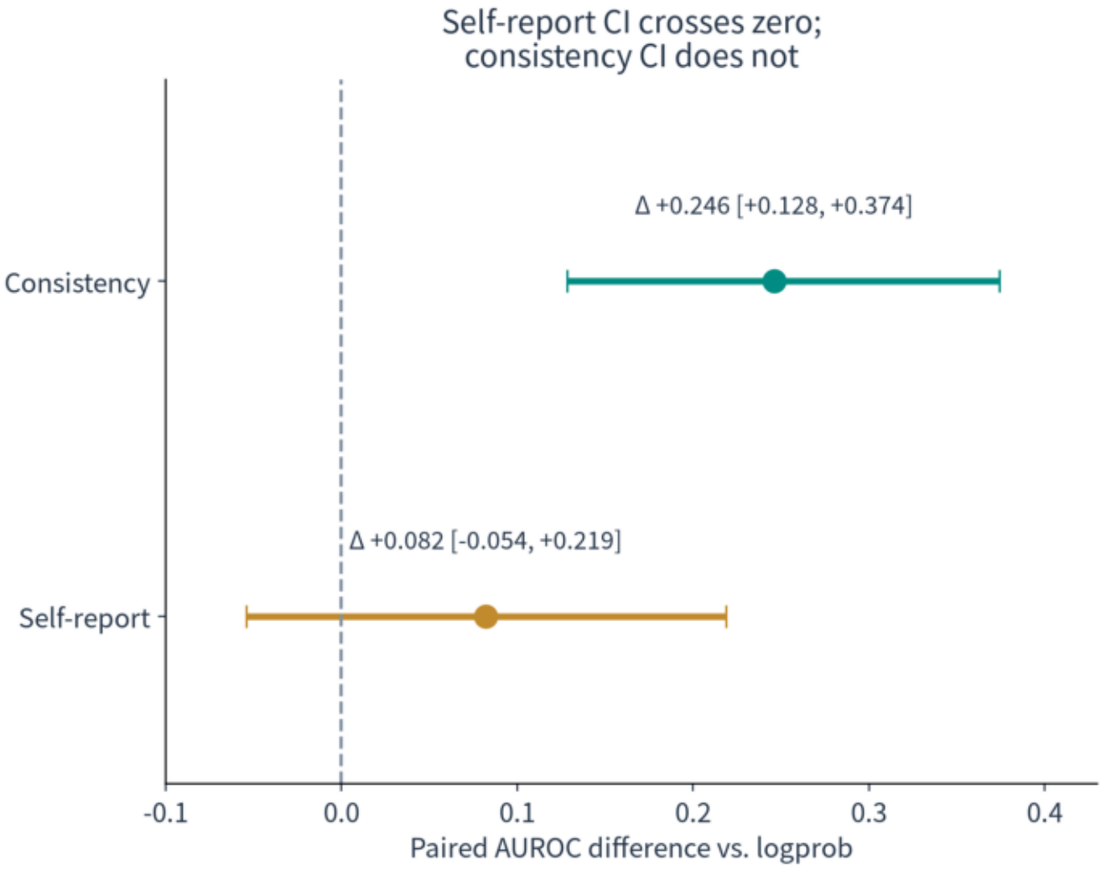
Sources: legacy_binary_v2/label_summary.json; new_signals_fast_v1/new_signal_metrics.json | As of 2026-10-20

24/40 incorrect calls invent missing inputs. Counts include retries within a task.

03 Consistency outperforms logprob on these legacy tasks

Same binary labels for all 198 steps; 1,000 task-cluster bootstrap replicates preserve within-task dependence.





After semantic deduplication (130 steps), consistency AUROC is 0.8793. The pre-mock-error paired CI crosses zero.
New scores use reconstructed contexts and no-thinking probes; legacy logprob uses thinking, so mechanisms are confounded.

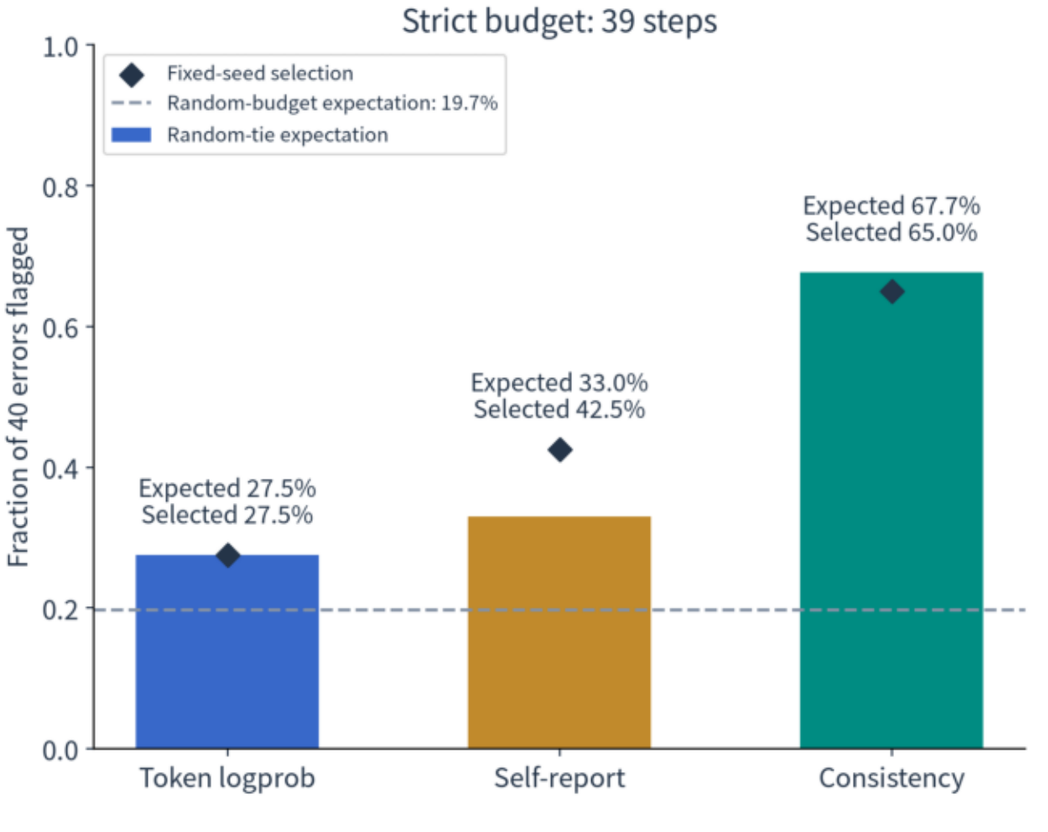
Sources: new_signals_fast_v1/new_signal_metrics.json / cohorts | As of 2026-10-20

The paired AUROC difference supports consistency over logprob;
the self-report advantage remains uncertain.

04 Low scores prioritize review, but scoring has a cost

Same strict budget: floor(0.2 × 198) = 39 steps (19.7%). Show random-tie expectations and fixed-seed selections separately.





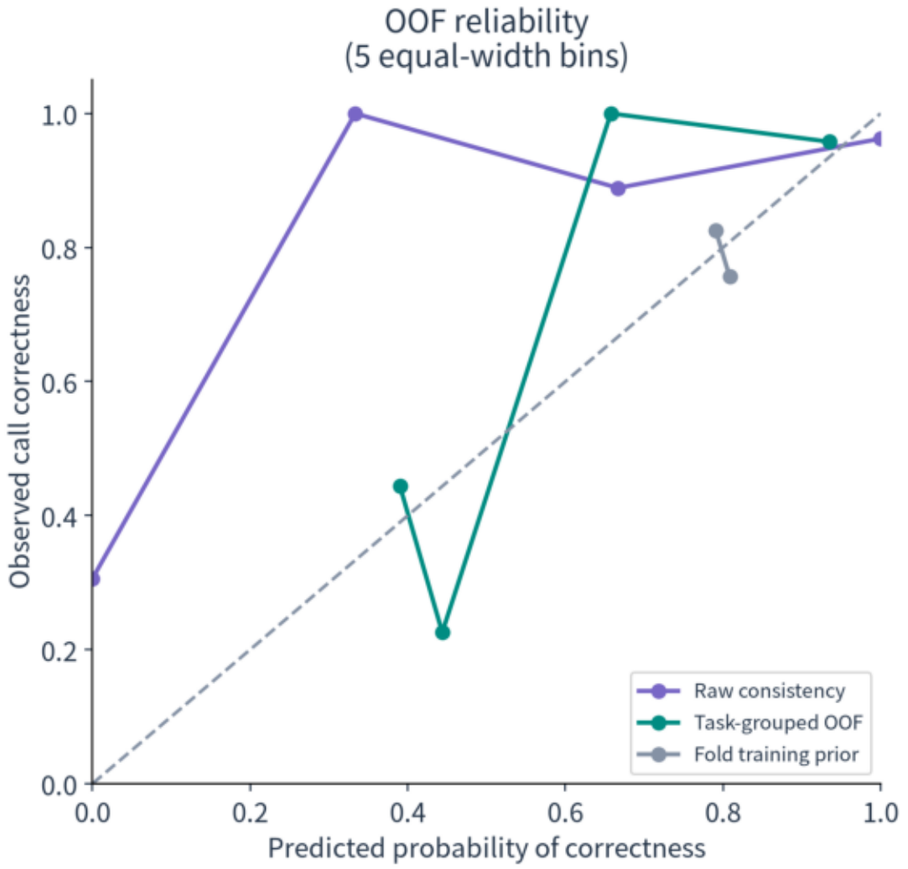
Consistency: the fixed seed selects 39 steps and flags 26/40 errors (65.0%); the random-tie expectation is 67.65%.
The earlier 68.69% used an expected budget of 39.6 steps. These are allocation simulations; no verification or correction ran.

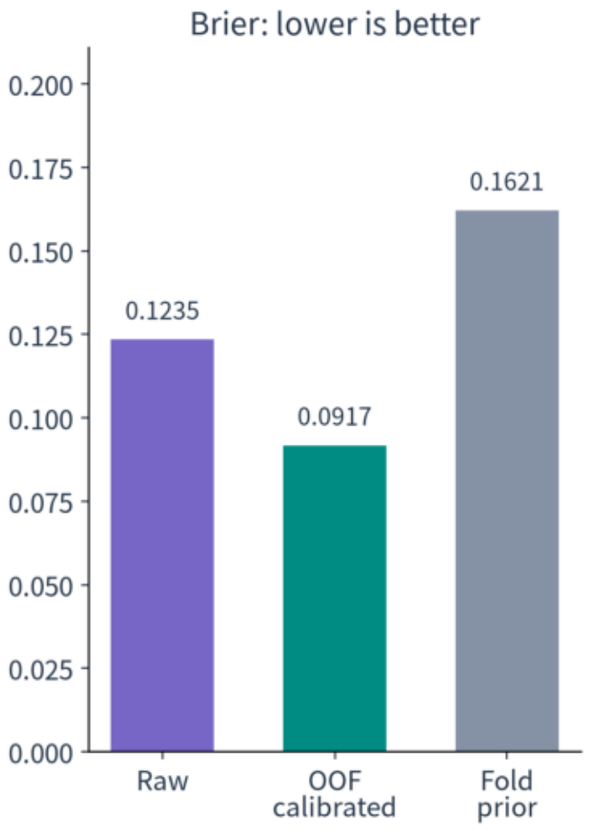
Sources: joined_scores.csv (frozen); data/strict_verification_budget.csv; calibration_metrics.json | As of 2026-10-20

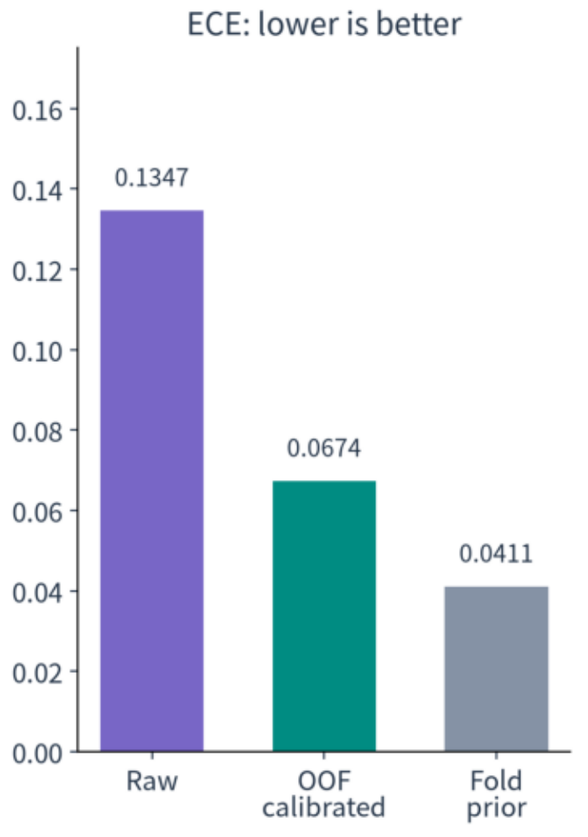
Use the same strict budget of 39/198 steps.
Separate random-tie expectations from fixed-seed selections.

05 Internal calibration improves probabilities, not ranking

Fixed five-fold StratifiedGroupKFold on 47 tasks with calls; one-variable sigmoid, C=1, no hyperparameter search.







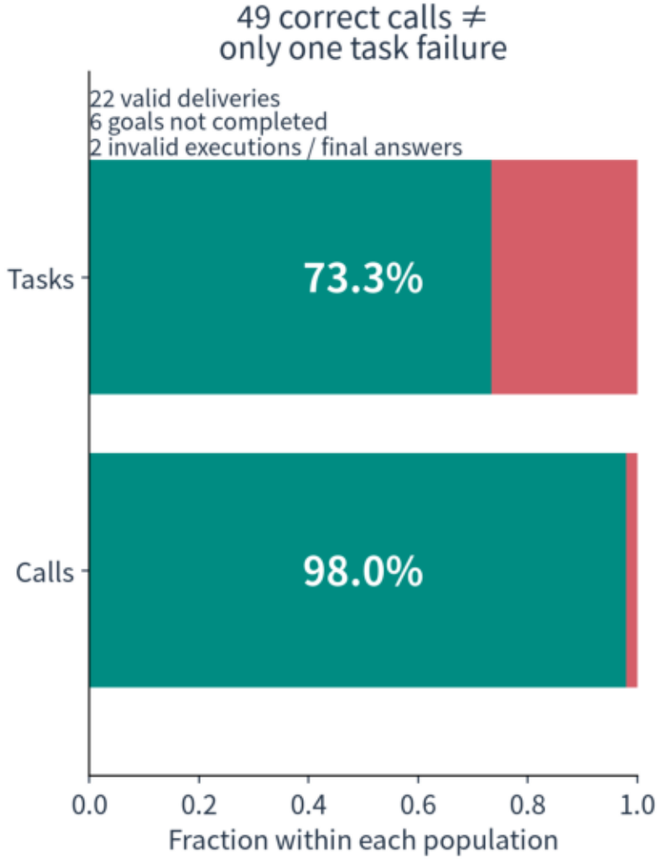
Brier: 0.1235 → 0.0917; ECE: 0.1347 → 0.0674, but prior ECE is lower (0.0411). ECE alone is insufficient.
OOF AUROC: 0.8812 → 0.8479; fold-specific maps change pooled ranks. A deployed positive-slope sigmoid preserves ranking.

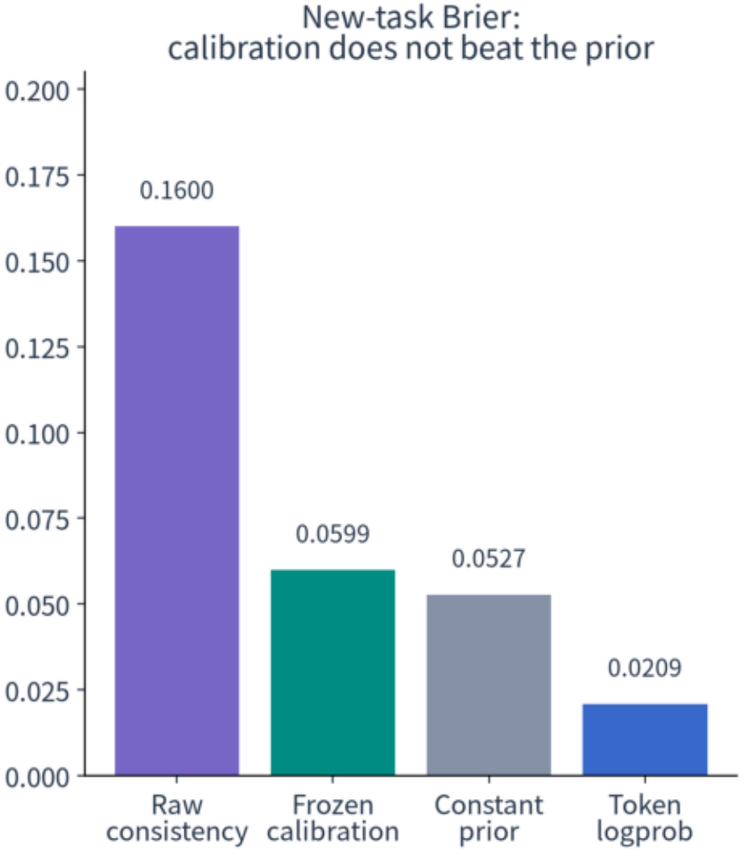
Sources: confidence_minimal_v1/calibration_metrics.json; internal evaluation, not an independent external test | As of 2026-10-20

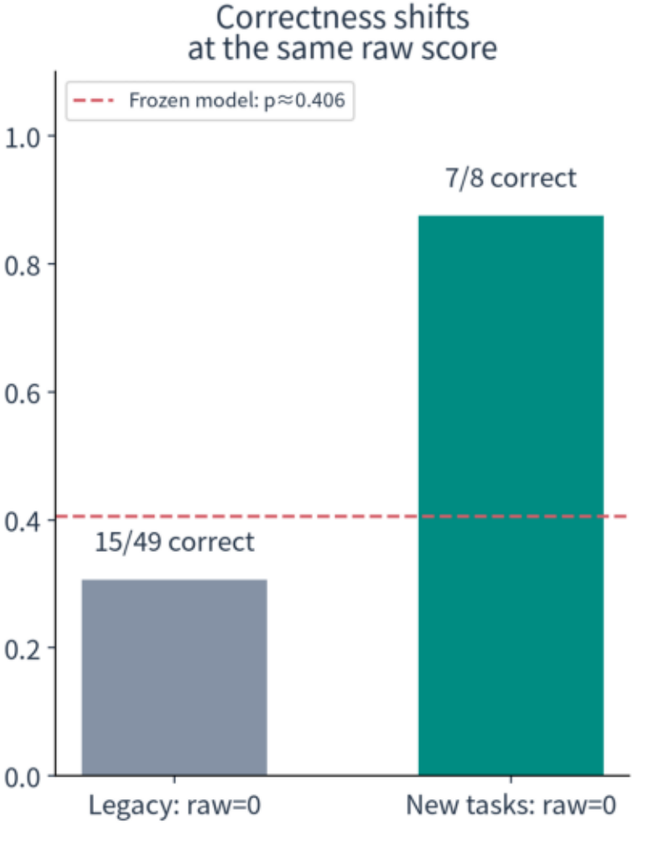
Task-grouped five-fold OOF reduces Brier error; the prior still has lower ECE.

06 New-task integration works; probability transfer is unproven

30 new tasks: 26 with calls and 4 without. With only one incorrect call, AUROC 0.9286 cannot establish stable generalization.







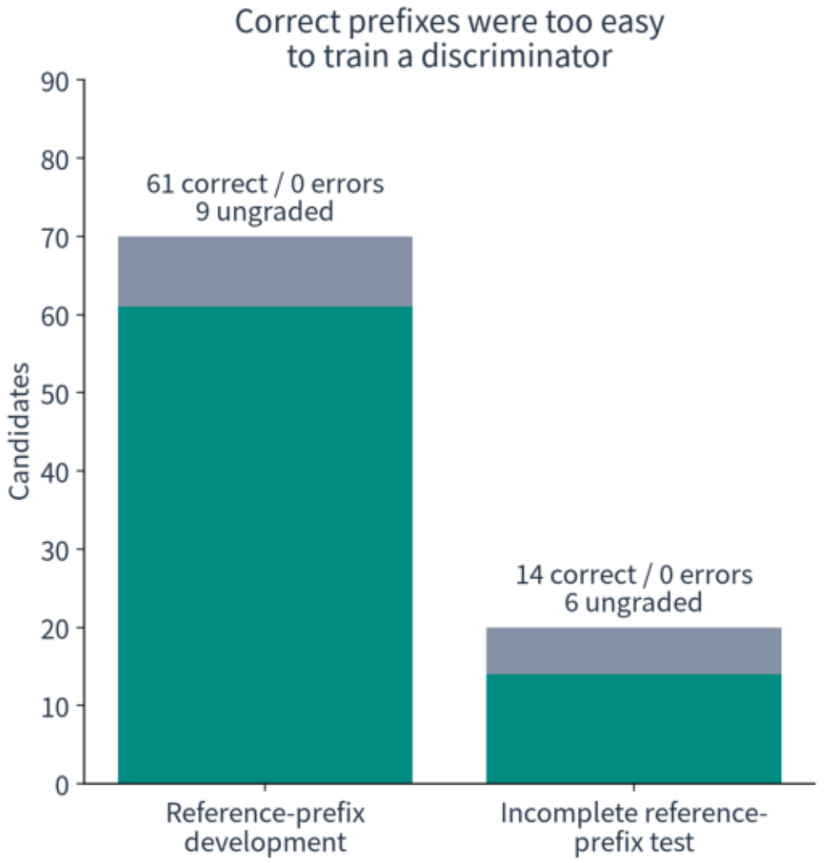
The rule flags 11/50 calls: the sole error plus 10 correct calls. Flagging 1/1 does not establish stable 100% detection.
At 98% call correctness, constant high scores can have low Brier. Success rates across different tasks do not prove capability gains.

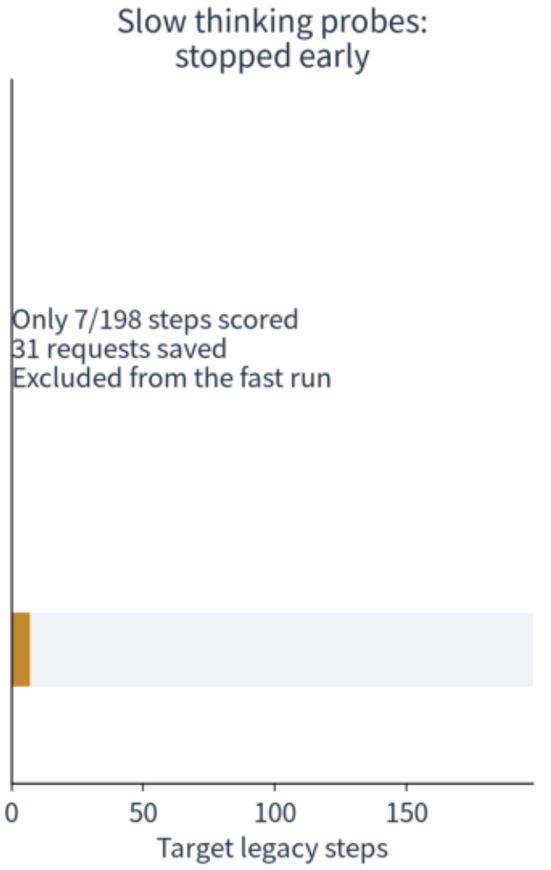
Sources: confidence_minimal_v1/validation_metrics.json; validation_task_outcomes.jsonl; frozen legacy step scores | As of 2026-10-20

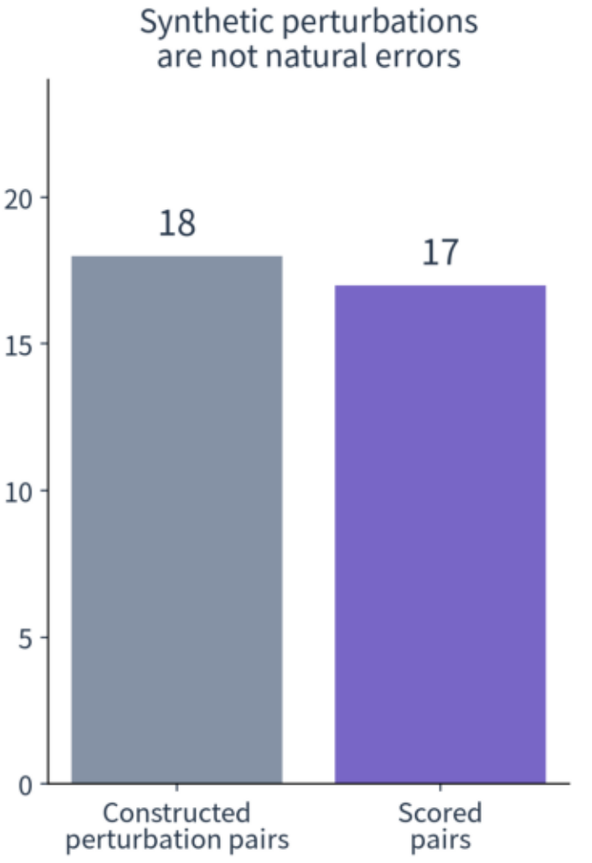
Only one incorrect call was observed. Calibrated Brier does not beat the prior.

07 Stopped experiments helped narrow the research scope

These diagnostics or incomplete pilots are not complete natural-task tests and must not be added to the main 198 steps.







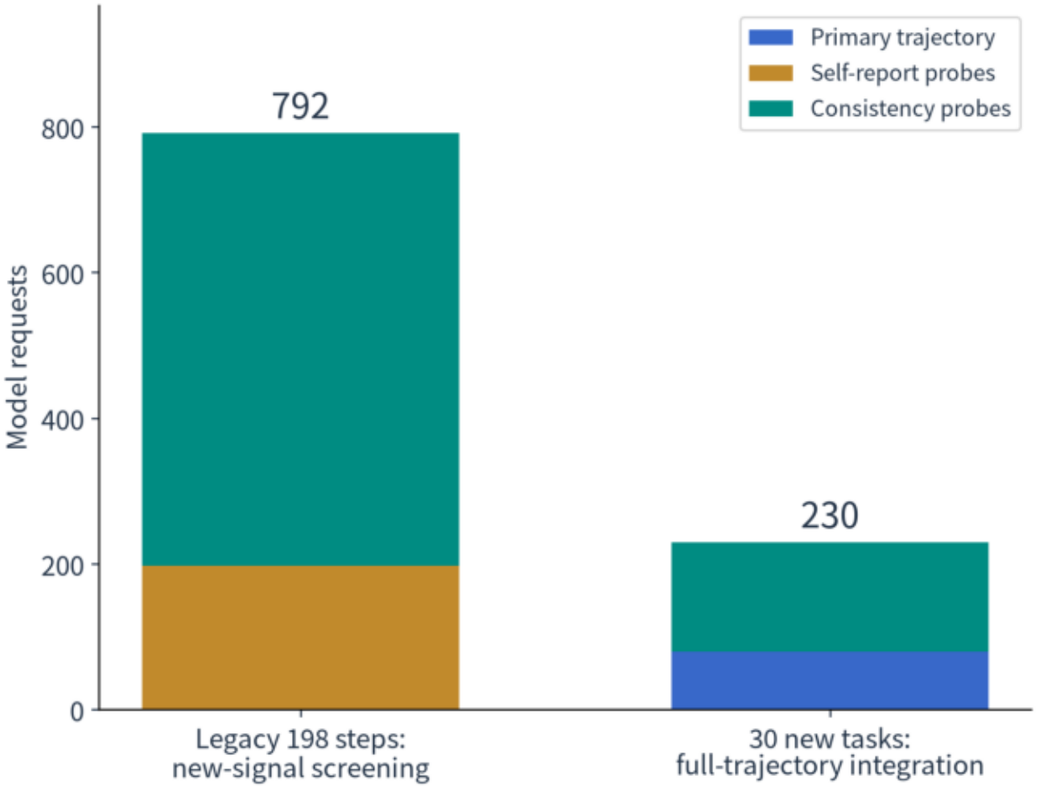
Reference-prefix tests covered 9/20 planned tasks and 20 candidates, with no negative examples for discrimination.
17 of 18 synthetic pairs were scored and fitted, with no independent stress holdout. Training fit does not establish natural-task generalization.

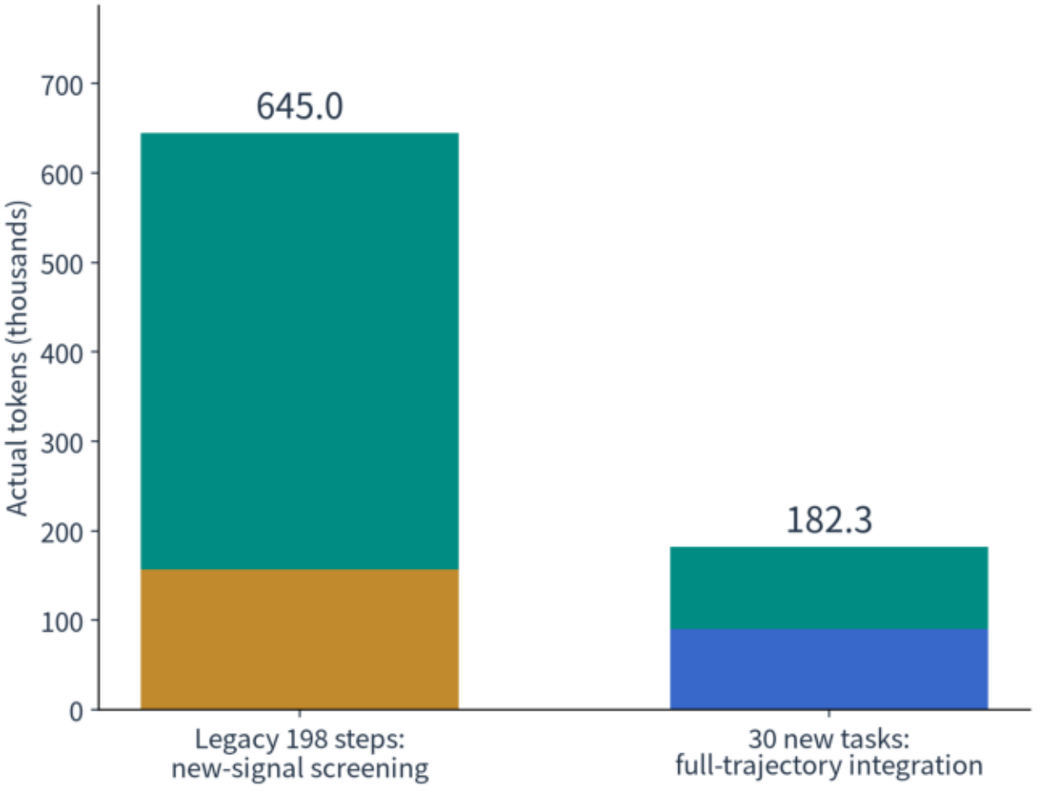
Sources: confidence_v1/prefix_diagnostic_summary.json; stress_frozen_spec.json; new_signals_v1/ | As of 2026-10-20

Reference prefixes lacked errors; thinking probes were slow;
synthetic stress data had no independent holdout evaluation.

08 Three probes cost extra: review budget is not inference cost

Show actual requests and tokens for each experiment. Different tasks and functions prevent direct efficiency comparisons.





New trajectories: 80 primary requests / 89,869 tokens + 150 probes / 92,383 tokens = 230 requests / 182,252 tokens.
All 50 candidates incur three probes before selection. Verification and correction were not run; their extra costs are not included.

Sources: new_signals_fast_v1/new_signal_metrics.json / collection_diagnostics.cost; validation_metrics.json / model_cost | As of 2026-10-20

Every new candidate incurs three probes. Probe tokens equal 102.8% of primary tokens.

09 A minimal prototype: pre-call score → probability → review flag

Qwen3-8B: thinking generates the candidate; three probes disable thinking. Current tool results and labels are excluded.



Mapping 0/3 agree → 40.6% | 1/3 agree → 66.7% | 2/3 agree → 85.5% | 3/3 agree → 94.5%

Implemented Full input logs; separate binary labels; task-grouped OOF; frozen model; new-task integration

To validate Confidence for no-tool endings; new errors from the same distribution; equal-cost correction gains

Next phase Validate reusable experience; update memory or policy; test whether recursive changes improve capability

Local call reliability is not task-success probability. Agreement can still be wrong. Sources: confidence_runtime.py; calibrator.json | 2026-10-20

Pre-call probabilities and review flags work.
Stopping confidence, actual correction, and experience updates remain untested.

下一步：先验证概率与检查的价值，再推进递归改进

1 数据与标签

保留自然连续轨迹，补足同分布错误；复核结束事件，避免 mock 覆盖限制充当错误真值。

2 置信度验证

冻结评分协议和校准器，在新的任务组检查 Brier、排序、覆盖率及错误类型。

3 预算实验

把评分与验证成本一起计入，与随机检查、全量检查比较实际纠错和成功率。

4 递归改进

只把独立核验有效的经验纳入更新，并保留未参与更新的任务检验能力提升。